

This is a postprint version of the following published document:

Astudillo, Armando; Al-Kaff, Abdulla; García, Fernando. (2022). Mono-DCNet: Monocular 3D Object Detection via Depth-based Centroid Refinement and Pose Estimation. *2022 IEEE Intelligent Vehicles Symposium (IV) [33rd IEEE IV]*, 4-9 June 2022, Aachen, Germany. IEEE. Pp.: 664-669.

DOI: <https://doi.org/10.1109/IV51971.2022.9827373>

©2022 IEEE. Personal use of this material is permitted. Permission from IEEE must be obtained for all other uses, in any current or future media, including reprinting/republishing this material for advertising or promotional purposes, creating new collective works, for resale or redistribution to servers or lists, or reuse of any copyrighted component of this work in other works.

Mono-DCNet: Monocular 3D Object Detection via Depth-based Centroid Refinement and Pose Estimation

Armando Astudillo¹, Fernando García¹, Abdulla Al-Kaff¹

Abstract—3D object detection is a well-known problem for autonomous systems. Most of the existing methods use sensor fusion techniques with Radar, LiDAR, and Cameras. However, one of the challenges is to estimate the 3D shape and location of the adjoining vehicles from a single monocular image without other 3D sensors; such as Radar or LiDAR. To solve the lack of the depth information, a novel method for 3D vehicle detection is presented. In this work, instead of using the whole depth map and the viewing angle (allocentric angle), only the depth mask of each object is used to refine the projected centroid and estimate its egocentric angle directly.

The performance of the proposed method is tested and validated using the KITTI dataset, obtaining similar results to other state-of-the-art methods for Monocular 3D Object Detection.

I. INTRODUCTION

Intelligent transportation systems are getting involve more and more in our lives. Developments in autonomous vehicles such as autonomous buses, delivery cars or taxi services [1][2][3][4] are already available. However, to accomplish the complex tasks associated to autonomous mobility, it is necessary to understand the environment surrounding the vehicle and the interactions with other road users. To achieve such an important task, there are several different methods to detect and locate objects in the 3D environment, most of them are based on the fusion of LiDAR sensors and cameras, so that image detectors are combined with the information from a 3D point cloud of one or several LiDAR in a 3D detector.

This kind of methods requires a very expensive and complex setup; since the LiDARs must be calibrated and synchronized with the cameras. Although there are automatic calibration methods [5][6], they usually demand manual refinement to get an accurate sensor fusion. All the aforementioned makes the 3D detection of objects with monocular cameras a field of high interest. Furthermore, the availability of 3D information based on a single camera provides enhanced information in systems with limited resources, or in situations where LiDAR sensors are not available. Even though the detection accuracy is lower than using LiDAR and camera fusion methods, the results in datasets like KITTI [7] provides adequate precision for certain applications.

This paper proposes a novel method of 3D Object Detection for monocular cameras based on the use of the object masks, gathered from a 2D instance segmentation network

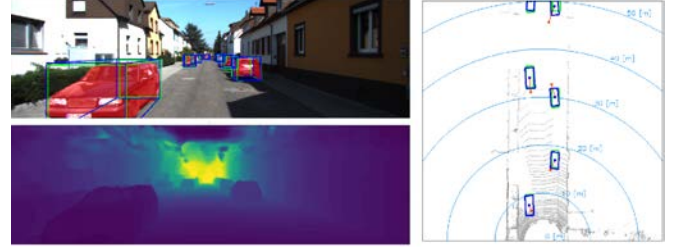


Fig. 1. Our proposed method for Monocular images. 2D masks are in red, 3D ground-truth in green, 3D centroid in orange and the final 3D estimation in blue.

and the estimated depth map obtained through a network working in parallel (Figure 1). Instead of relying on the model to learn the depth of the object's centre with only the RGB image, we use a network that estimates the depth of the entire scene, this allows it to be used in a wide variety of applications, such as in the estimation of an occupancy map or in SLAM applications [8][9][10].

The main contributions of the proposed work are:

- The use of 2D detection masks instead of boxes, avoiding the uncertainty with the background in the depth image.
- The use of the depth mask instead of the projection of it in 3D points, saving execution time, computational consumption and maintaining the information consistency in each region.
- To the best of our knowledge, this is the first method for Monocular 3D Object Detection that estimates directly the egocentric angle, instead of calculating this angle by estimating the allocentric angle, as it is common in monocular methods.
- The use of a hybrid angle estimation based on a classification and a regression.

II. RELATED WORK

3D Object Detection techniques for intelligent systems is discussed on this section. It can be classified in: methods based on sensor fusion between LiDAR and Camera, point cloud methods and image-based methods.

Fusion-based 3D detection: These methods rely in different types of architectures to join the best of both sensors: the ability to classify the objects in image and accuracy of the 3D point clouds [11][12]. This configuration allows to be adapted to some Monocular-based methods referred as pseudo-LiDAR. Pseudo-LiDAR methods aim to refine the depth maps estimated through the neural networks by projecting them into a point cloud, allowing sensory fusion

¹Authors are with the Autonomous Mobility and Perception Lab (AMPL), Universidad Carlos III de Madrid, Spain {aastudil, fegarcia, akaff}@ing.uc3m.es

We gratefully acknowledge the support of NVIDIA Corporation with the donation of the GPUs used for this research.

methods to be applied and taking advantage of their accuracy despite their high computational cost [13][14].

LiDAR-based 3D Detection: Modern LiDAR sensors are able to provide a highly dense and accurate point cloud, this makes LiDAR-based detection method a very attractive field for researchers and practitioners, despite of the high costs of these devices. Many methods have been introduced for LiDAR-based object detection: some propose to voxelize the point cloud [15][16][17], while others use the point cloud image created by projecting the points on the ground plane, i.e. birds-eye-view (BEV), and use modified 2D image detection methods [18][19].

Monocular-based 3D Detection: The difficulty in estimating size and depth in an RGB Monocular image means that many of the works developed are based on geometric assumptions [20][14][21][22] or by comparing the projection of 3D models or key-points with instances of the objects in the image [23][24]. Other methods use RGB image to estimate the depth and dimensions of each object independently from the rest of the scene [25][26]. In this work, we propose the use of an intermediate network that estimates the depth of a complete scene. This configuration is compatible with different applications which take advantage of the use of 3D information. In [27] authors propose the use of two parallel networks that share information between the RGB image and the depth map by modifying the weights of the filters to obtain the object's pose.

The main similarity of these monocular-based methods is the estimation of the allocentric angle of the object, that is to say, the viewing angle of the object in Image Coordinates. This estimation is invariant as far as the shape of the object is concerned, and, in order to calculate the egocentric angle, it must be triangulated with respect to the allocentric angle and the 3D location, which increase the error dimension of the calculus by propagating previous errors. However, we suggest that the egocentric angle is inherently found in the depth map of the object, so it is not necessary to post-process the allocentric angle to find it, but it can be directly estimated.

III. METHODOLOGY

The proposed approach is composed of three fundamental parts, connected as shown in Figure 2: The depth estimation of the complete scene, the 2D image detection and the estimation of the 3D pose of each detected object.

A. Depth Estimation

Nowadays, there are several methods to estimate the depth from RGB images. These can vary depending on whether a sequence of images, a single image, or stereo images are used. In this case, a single image methodology has been chosen in order to infer faster. In most cases, the depth of the object is estimated per instance ("instance-centric") [22][14], however, for this proposal, the whole scene's depth is estimated, then, each object is refined. This refinement helps to reduce the errors originated due to the steep gradients for each object, which cause that the projected centroid from

the depth map may have an error of several meters as it can be appreciated in Figure 1 and Figure 5.

Depth estimation of the scene from a simple RGB image is based on DattNet, as it has a better trade-off between accuracy and speed. Table I shows a comparison of its performance with other similar methods [28][29][30]. DattNet is based on a modified Encoder-Decoder CNN architecture without a dense feature extraction stage. Instead of this layer, authors proposed to use Attention Layers that are faster without losing critical environment information, obtaining similar precision but three times faster than BTS [28].

TABLE I
COMPARATIVE RESULTS FOR THE ABSOLUTE RELATIVE ERROR FOR
DEPTH ESTIMATION.

Method	Absolute Relative Error	Inference Time [ms]
monoDepth2	0.106	N/A
DORN	0.072	500
BTS	0.060	60
DattNet	0.064	20

B. 2D Detector

In parallel to the depth estimation stage, an instance detector is performed, providing 2D bounding boxes, masks and classes of the surrounding obstacles.

Using instance segmentation methods allows to separate the background of the scene from the object, avoiding to create complex models. For this reason, Detectron2 framework [31] is used.

C. 3D Estimation

The 3D estimation stage fuses the information of the two branches presented before and provides the final estimation of the pose of the obstacles.

Data Processing. The first step of this phase is to process the output of the two previous networks: the scene depth estimation and the 2D object detection. This is done by obtaining the depth information within the area of the image indicated by each mask provided by Detectron2. This step is straightforward, as the output of both branches have the same dimensions, if a stereo-based method were used it would require to use accurate extrinsic calibration. In the second step, the average value of every object's depth mask is calculated to determine the depth of the centroid ('Z'). Then, from the 'Z' value, the x-y image point and the 2D bounding box, it is possible to obtain the X and Y position of the inferred centre (see equations (1),(2),(3)) as shown in Figure 3. This process reduces the execution time and computational consumption in comparison to projecting the whole depth scene into 3D coordinates. The centroid obtained contains errors due to possible occlusion which may displace the projected centre, in addition to the errors inherent to the depth estimation. Therefore, the 3D refinement module included in this work incorporates a branch that calculates the offset between the

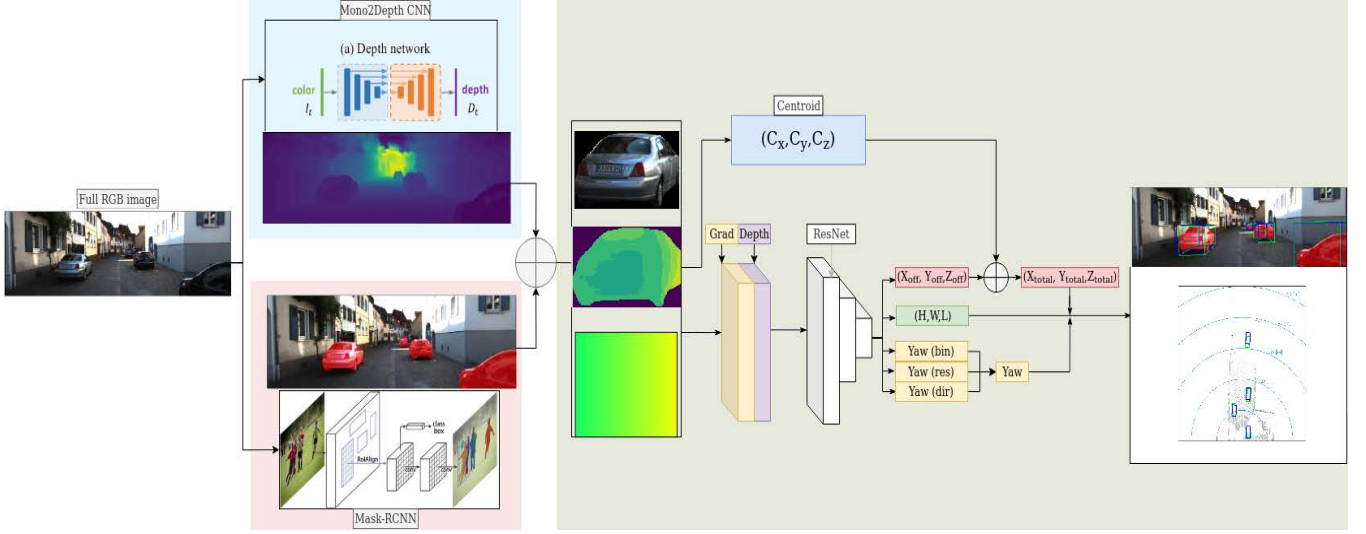


Fig. 2. Proposed method for monocular 3D Detection.

real and the proposed 3D centroid and learns its regression.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} \quad (1)$$

$$u = \frac{u'}{w'} \quad (2)$$

$$v = \frac{v'}{w'} \quad (3)$$

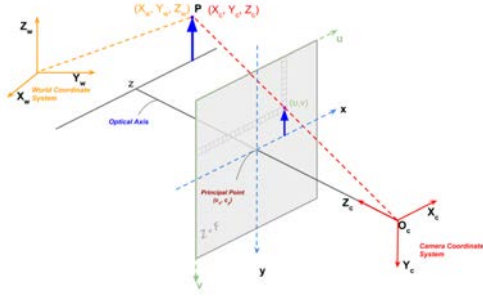


Fig. 3. Representation between the 2D world and 3D world.

Egocentric angle and hybrid estimation. Most of the works related to Monocular 3D Detection estimate the egocentric angle of the object from the allocentric angle and the 3D position of the object. The reason for this is that the visual aspect of an object in the 2D RGB image that is placed in different positions but with the same egocentric angle, is different. This oddity causes that the models do not know how to estimate efficiently the egocentric angle. On the other hand, the allocentric angle does not vary visually and it is easier to estimate. The calculation of the egocentric angle (yaw) depends on the 3D position ($\tilde{X}, \tilde{Z}$) and the allocentric angle (α) as shown by equations (4),(5). Therefore, existing methods learn the allocentric pose of the object and then calculate the egocentric angle. We propose to eliminate the potential error due to high depth gradients and to directly

estimate the egocentric angle. The egocentric angle can be inferred straight from the depth mask representation of the object, without the background, since this information is present in the depth mask of the object. The results obtained demonstrate the reliability of our hypothesis. Figure 4 represents a more graphic explanation of the difference between the allocentric angle and the egocentric angle.

$$\alpha = yaw - \theta \quad (4)$$

$$\theta = \arctan \left(\frac{\tilde{X}}{\tilde{Z}} \right) \quad (5)$$

In order to learn the egocentric angle, an angle normalisation is proposed. The angle is normalised between -90 and 90, instead of -180 and 180, since the depth gradients for opposite 3D boxes are the same. For egocentric angle estimation, we propose to use a hybrid method, similar to the one proposed in [19], which consist in the classification in bins of the angle to which the object belongs, and the regression of an offset that allows to reduce the error produced by the angular resolution caused by the bins. In addition to this estimation, we include an orientation classifier to cover a 360 degree.

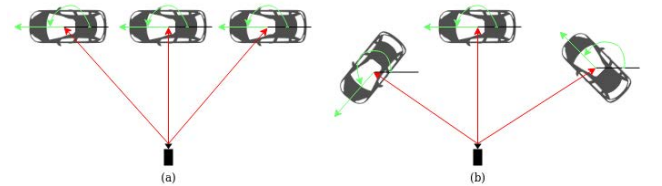


Fig. 4. In (a), the egocentric angle of the car, in green, are all facing the left, but the allocentric orientation and viewpoint will change when the car moves from the right to the left. In (b), the egocentric angle of the car differ, but both the allocentric angle in the camera coordinates and the viewpoint remain unchanged.

3D Pose refinement. The 3D pose refinement module takes as input the depth region given by its mask and the lateral

gradient. This gradient has a variation of values from right to left and helps the model to differentiate in a simple way the objects located at the left or right of the scene that have similar viewpoint.

The backbone chosen to extract the features of the object is a ResNet-50 [32] since it has the best trade-off between parameter size and results (see ablation studies section: IV-E). The output of the ResNet is modified in such a way that it extracts the features without the classification stage, this corresponds to the output heads present on the right side of the Figure 2. We define a head for each parameter determining the 3D pose: an offset in the 3D location ($X_{offset}, Y_{offset}, Z_{offset}$), the absolute dimensions of the object (H, W, L), the bin of the angle to which it belongs (yaw_{bin}), the residual of that angle (yaw_{res}) and the orientation of the box (yaw_{dir}).

The output of the location is added to the proposed centroid (C_X, C_Y, C_Z) increasing its accuracy as the network learns to correct the errors in the resolution of the estimation of the whole depth map (6). The egocentric angle is reconstructed from the estimation of the bin to which it belongs, the residual applied and the orientation of the box. As we are using the mask of depth of the object, the absolute dimension of the object is also codified in it. This allows to directly provide the size of the each object, instead of the residual value to the average size per class as it is done in other works.

$$\begin{aligned}\tilde{X} &= C_X + X_{offset} \\ \tilde{Y} &= C_Y + Y_{offset} \\ \tilde{Z} &= C_Z + Z_{offset}\end{aligned}\quad (6)$$

Losses. A multi-task loss (equation 7) is proposed for the learning stage. It is composed by: the 3D location loss (L_{loc}), the global dimension loss (L_{dim}) and the egocentric angle loss (L_{yaw}).

$$L_{total} = L_{loc} + L_{dim} + L_{yaw} \quad (7)$$

The location loss is a weighted regression between the refined centroid and the centre of the 3D ground truth.

$$\begin{aligned}L_{loc} &= 1.0 * SmoothL1(\tilde{X}, X_{gt}) \\ &+ 0.8 * SmoothL1(\tilde{Y}, Y_{gt}) \\ &+ 1.2 * SmoothL1(\tilde{Z}, Z_{gt})\end{aligned}\quad (8)$$

To estimate the dimensions a regression between the dimension head's logits and the 3D dimensions of the ground truth is used.

$$\begin{aligned}L_{dim} &= SmoothL1(H, H_{gt}) \\ &+ SmoothL1(W, W_{gt}) \\ &+ SmoothL1(L, L_{gt})\end{aligned}\quad (9)$$

The egocentric bin is an eight-bin classification as in [20]. The residual of the egocentric angle is a regression between the egocentric angle of the ground truth and the computation

of its bin. The orientation of the box is a binary classification.

$$\begin{aligned}target_{yaw} &= yaw_{gt.discrete} - yaw_{gt} \\ L_{yaw} &= SmoothL1(yaw_{res}, target_{yaw}) \\ &+ CE(yaw_{bin}, yaw_{gt.discrete}) \\ &+ CE(\tilde{dir}, dir_{gt})\end{aligned}\quad (10)$$

The *SmoothL1* loss is used for regressions, while the *CrossEntropyLoss* (*CE*) is used for classifications.

IV. EXPERIMENTAL RESULTS

In this section we present the quantitative results in comparison with other Monocular methods, as well as the qualitative results of our approach.

A. Test Configuration

For training and evaluation of 3D detectors, both in image and LiDAR, the KITTI dataset is widely used. This dataset consists of 7,481 frames for training and 7,518 for testing. Each frame is composed by an RGB image and a point cloud captured with a 64-layers LiDAR. It also contains the intrinsic calibration files of the camera as well as the extrinsic calibration between camera and LiDAR. For the training stage of the 3D detection networks, the split presented in [33] is used. It contains 3,712 training frames and 3,769 validation frames.

The results are determined using Precision-recall curves with an IoU of 0.7 for cars, which is the class to be evaluated in the presented work. Currently, a 40 points Average Precision ($AP|_{R_{40}}$) is used instead of the $AP|_{R_{11}}$ used in the past. However, since the existing methods use this old interpolation, we will provide the results with this metric.

B. Experiment Setup

The model has been developed using the Deep Learning Pytorch framework [34]. A fine-tuning of the ResNet-based backbone is performed, this backbone is pre-trained in ImageNet [35]. For the new convolutional layers a Xavier initialisation is used, while the classifiers are initialised with a uniform Kaiming methodology.

Unlike other methods that use ground-truth to generate new samples, we do not apply any type of data-augmentation.

The network has been optimised by Stochastic Gradient Descent (SGD) with a fixed learning rate of 0.01, as this offered the best performance for learning the egocentric angle.

C. Quantitative Results

Table II shows the results for the validation split from KITTI dataset for 3D Detection and BEV, as this is the common evaluation splits presented in the works developed for Monocular 3D Detection. In this table, it can be appreciated that the presented method (in green), despite being smaller and less computationally demanding than those based on pseudo-LiDAR, is able to obtain results similar to these ones, and outperform in several metrics the classic method based on monocular only with RGB image (in orange).

TABLE II
VALIDATION RESULTS FOR CAR 3D LOCALIZATION AND BEV DETECTION ON THE KITTI VAL SPLIT WITH 0.7 OF IOU.

METHOD	AP 3D Detection			AP BEV Detection			Inference Time [seconds]
	Easy	Moderate	Hard	Easy	Moderate	Hard	
Mono3D	6.55	5.19	4.10	5.22	5.19	4.13	-
Deep3DBox	5.85	4.10	3.84	9.99	7.71	5.3	-
MF3D	10.53	5.69	5.39	22.03	13.63	11.6	0.03
MonoPSR	12.75	11.48	8.59	20.63	18.67	14.45	0.20
MonoGRNet	13.88	10.19	7.62	-	-	-	0.04
MonoFENet	17.54	11.16	9.94	30.21	20.47	17.58	0.15
MonoDIS	18.05	14.98	13.42	24.26	18.43	16.95	0.10
Pseudo-LiDAR	19.50	17.20	16.20	33.70	24.60	20.10	0.40
Ours	19.17	14.77	12.28	30.05	21.54	17.78	0.13

The accuracy provided by the method presented is stable in terms of BEV and 3D, unlike the other methods included in the comparison. The methods that provides better performance in BEV do not have good results in 3D and vice versa. Thus, our hypothesis concerning the fact that the egocentric angle is encoded within the depth mask of the object is demonstrated.

Another aspect to be mentioned is that many training samples have been discarded, so the accuracy of our method does not reach its maximum. This is due to the fact that the KITTI dataset provides a 2D bounding box and not a mask, so the matching between proposal and ground-truth is difficult to compute in certain cases, due to occlusions or truncations.

D. Qualitative Results

The Figure 5 shows the performance of the proposed method in some test scenes. It can be seen how these results are accurate enough for 3D car detection, proving the efficiency and the usability of the proposal.

E. Ablation Studies

Several configurations were tested in order to achieve the best trade-off between 3D and BEV accuracy. Table III shows that the best configuration is based on ResNet-50 plus a combination of seventeen bins for the angle and a classifier for orientation.

V. CONCLUSIONS

In this paper, a new method for Monocular 3D Detection is presented. This approach is based on the use of an instance segmentation network and a full scene depth estimation network, which in combination with the 3D regression module is able to obtain the poses of the objects without a high power consumption and precisely. The estimation of the egocentric angle of the object in a straightforward way is also proposed for the first time in Monocular methods and not through the estimation of the allocentric angle. The angle estimation is composed by three different heads: one based on bins classification, a regressor of the residual and an orientation estimator.

The results obtained validate the performance of the model, achieving an accuracy similar to current state-of-the-art methods based completely on Monocular imaging.

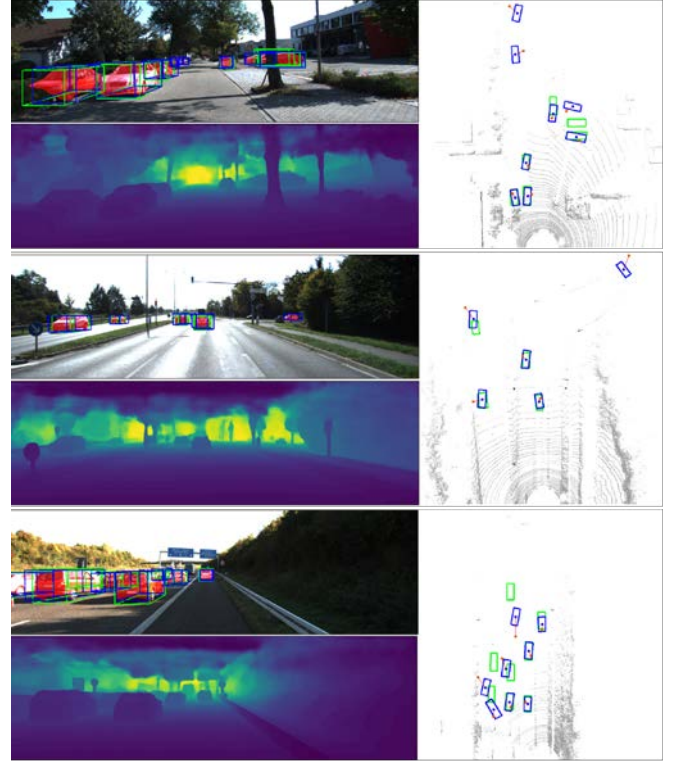


Fig. 5. Qualitative results for KITTI Dataset. Images are composed by the RGB Image, the estimated depth and the BEV representation. Green boxes represent the ground truth and blue boxes represent the output of the network.

The ongoing development of new datasets with instance segmentation and 3D boxes will allow this system to learn more efficiently and achieve optimal results.

ACKNOWLEDGEMENTS

This work has been supported by the Madrid Government (Comunidad de Madrid-Spain) under the Multianual Agreement with UC3M (“Fostering Young Doctors Research”, APBI-CM-UC3M), and in the context of the V PRICIT (Research and Technological Innovation Regional Programme).

Also, We gratefully acknowledge the support of NVIDIA Corporation with the donation of the GPUs used for this



TABLE III
ABLATION RESULTS FOR CAR 3D LOCALISATION AND BEV DETECTION ON THE KITTI VAL SPLIT WITH 0.7 OF IOU.

Backbone	Angle Target	Number of bins	Orientation Classifier	AP 3D Detection			AP BEV Detection		
				Easy	Moderate	Hard	Easy	Moderate	Hard
ResNet-18	360°	16	No	4.42	10.56	10.19	7.44	12.63	11.14
ResNet-50	360°	16	No	9.72	13.56	13.11	18.80	19.07	16.16
ResNet-50	180°	16	No	14.95	13.32	11.48	23.27	18.73	16.06
ResNet-50	180°	16	Si	19.17	14.77	12.28	30.05	21.54	17.78
ResNet-50	180°	8	Si	19.24	15.20	14.14	29.53	21.36	17.69

research.

REFERENCES

- [1] J. Beltrán, C. Guindel, I. Cortés, A. Barrera, A. Astudillo, J. Urdiales, M. Álvarez, F. Bekka, V. Milanés, and F. García, "Towards autonomous driving: a multi-modal 360° perception proposal," in *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 2020, pp. 1–6.
- [2] B. Yang, W. Luo, and R. Urtasun, "Pixor: Real-time 3d object detection from point clouds," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7652–7660.
- [3] M. A. de Miguel, F. M. Moreno, F. García, J. Armingol, and R. Martín, *Autonomous Vehicle Architecture for High Automation*, 04 2020, pp. 145–152.
- [4] D. Martín Gómez, P. Marín, A. Hussein, A. de la Escalera, and J. Armingol, *ROS-based Architecture for Autonomous Intelligent Campus Automobile (iCab)*, 01 2016, pp. 257–272.
- [5] J. Beltrán, C. Guindel, and F. García, "Automatic extrinsic calibration method for lidar and camera sensor setups," 2021.
- [6] C. Guindel, J. Beltrán, D. Martín, and F. García, "Automatic extrinsic calibration for lidar-stereo vehicle sensor setups," in *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*, 2017, pp. 1–6.
- [7] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the KITTI Vision Benchmark Suite," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3354–3361.
- [8] Y. Li, C. Xie, H. Lu, X. Chen, J. Xiao, and H. Zhang, "Scale-aware monocular slam based on convolutional neural network," in *2018 IEEE International Conference on Information and Automation (ICIA)*, 2018, pp. 51–56.
- [9] C. Zhao, L. Sun, P. Purkait, T. Duckett, and R. Stolkin, "Learning monocular visual odometry with dense 3d mapping from dense 3d flow," *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 6864–6871, 2018.
- [10] S. Y. Loo, A. Amiri, S. Mashohor, S. Tang, and H. Zhang, "Cnn-svo: Improving the mapping in semi-direct visual odometry using single-image depth prediction," 05 2019, pp. 5218–5223.
- [11] C. R. Qi, W. Liu, C. Wu, H. Su, and L. J. Guibas, "Frustum PointNets for 3D Object Detection from RGB-D Data," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 918–927.
- [12] J. Ku, M. Mozifian, J. Lee, A. Harakeh, and S. L. Waslander, "Joint 3d proposal generation and object detection from view aggregation," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.
- [13] Y. Wang, W. Chao, D. Garg, B. Hariharan, M. Campbell, and K. Q. Weinberger, "Pseudo-lidar from visual depth estimation: Bridging the gap in 3d object detection for autonomous driving," in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 8437–8445.
- [14] W. Bao, B. Xu, and Z. Chen, "Monofenet: Monocular 3d object detection with feature enhancement networks," *IEEE Transactions on Image Processing*, vol. 29, pp. 2753–2765, 2020.
- [15] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, 2018.
- [16] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, "PointPillars: Fast encoders for object detection from point clouds," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 12 697–12 705.
- [17] S. Shi, Z. Wang, J. Shi, X. Wang, and H. Li, "From points to parts: 3d object detection from point cloud with part-aware and part-aggregation network," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PP, pp. 1–1, 02 2020.
- [18] J. Beltrán, C. Guindel, F. M. Moreno, D. Cruzado, F. García, and A. de la Escalera, "BirdNet: a 3D object detection framework from LiDAR information," in *Proc. IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2018, pp. 3517–3523.
- [19] A. Barrera, C. Guindel, J. Beltrán, and F. García, "Birdnet+: End-to-end 3d object detection in lidar bird's eye view," 2020.
- [20] A. Mousavian, D. Anguelov, J. Flynn, and J. Košecká, "3d bounding box estimation using deep learning and geometry," 12 2016.
- [21] T. He and S. Soatto, "Mono3d+: Monocular 3d vehicle detection with two-scale 3d hypotheses and task priors," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 8409–8416, 07 2019.
- [22] J. Ku, A. Pon, and S. Waslander, "Monocular 3d object detection leveraging accurate proposals and shape reconstruction," 06 2019, pp. 11 859–11 868.
- [23] A. Kundu, Y. Li, and J. M. Rehg, "3d-rnn: Instance-level 3d object reconstruction via render-and-compare," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 3559–3568.
- [24] C. Li, Q.-H. Tran, X. Yu, and M. Chandraker, "Deep supervision with shape concepts for occlusion-aware 3d object parsing," 07 2017.
- [25] A. Simonelli, S. R. Bulò, L. Porzi, M. Lopez-Antequera, and P. Kontschieder, "Disentangling monocular 3d object detection," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 1991–1999.
- [26] Z. Qin, J. Wang, and Y. Lu, "Monogonet: A geometric reasoning network for monocular 3d object localization," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, pp. 8851–8858, 07 2019.
- [27] M. ng, Y. Huo, H. Yi, Z. Wang, J. Shi, Z. Lu, and P. Luo, "Learning depth-guided convolutions for monocular 3d object detection," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 4306–4315.
- [28] J. H. Lee, M.-K. Han, D. W. Ko, and I. H. Suh, "From Big to Small: Multi-Scale Local Planar Guidance for Monocular Depth Estimation," *arXiv e-prints*, p. arXiv:1907.10326, Jul. 2019.
- [29] C. Godard, O. M. Aodha, and G. J. Brostow, "Digging into self-supervised monocular depth estimation," *CoRR*, vol. abs/1806.01260, 2018. [Online]. Available: <http://arxiv.org/abs/1806.01260>
- [30] H. Fu, M. Gong, C. Wang, K. Batmanghelich, and D. Tao, "Deep ordinal regression network for monocular depth estimation," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2002–2011.
- [31] Y. Wu, A. Kirillov, F. Massa, W.-Y. Lo, and R. Girshick, "Detectron2," <https://github.com/facebookresearch/detectron2>, 2019.
- [32] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [33] X. Chen, K. Kundu, Y. Zhu, H. Ma, S. Fidler, and R. Urtasun, "3d object proposals using stereo imagery for accurate object class detection," in *TPAMI*, 2017.
- [34] A. e. a. Paszke, "Pytorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems 32*. Curran Associates, Inc., 2019.
- [35] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 248–255.